

Early programming methods

Modern programmers write code using human-readable text. The first programmers, on the other hand, wrote code in 0s and 1s, the language of the computer.

History of punch cards

Before program instructions could be stored on disks or magnetic tape, they were stored on punch cards. Programmers punched sequences of holes into stiff pieces of paper and then fed them into the computer to run a program. The design of punch cards gradually became more and more sophisticated.

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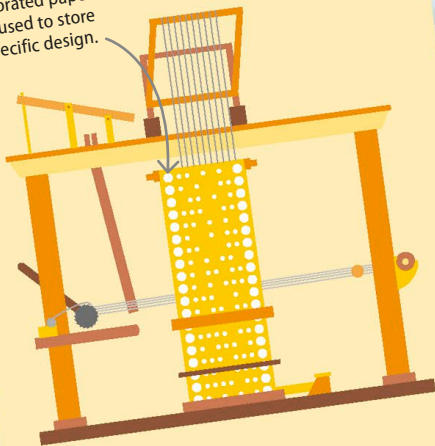
◀ 38–39 The computer chip

◀ 80–81 Bits and digitization

“At each increase of **knowledge...**
human labour becomes **abridged.**”

Charles Babbage (1791–1871),
English mathematician and inventor

Perforated paper was used to store a specific design.



1725: Basile Bouchon

In Lyon, France, textile worker Basile Bouchon created a method to store weaving patterns in a piece of tape. Where there was a hole in the tape, the needle on the loom stayed still. If there was no hole, the needle was pushed forward and the thread was lifted. Instead of trying to memorize complicated patterns and to avoid mistakes, weavers simply shifted the tape up and down. His creation was the first semi-automated industrial machine.

1822: The Difference Engine

English mathematician Charles Babbage was tired of typographical errors in his books of mathematical tables. These books had lists of pre-computed numbers, which were used in navigation, astronomy, and statistics. Babbage drew up the design for the Difference Engine, a mechanical calculator that could produce these tables automatically. While his design was good, the engine was very expensive to make.

The Difference Engine used columns of gears to make its calculations.

